

Standard Operating Procedure for taking skin biopsies used at Imperial College London and associated hospitals.

Prepared by Harriet Kemp, 5th February 2024

The procedure must only be performed by qualified health professionals who have undergone specific training in skin biopsy collection.

Biopsy Procedure

1. Confirm patient consent and check no allergies to local anaesthetic and no anti-coagulants
2. Patient to be lying on examination couch with lower leg exposed
3. Site of biopsy marked at 10cm proximally from lateral malleolus on the lower leg
4. Hand washing and drying as per full aseptic technique and use of sterile gloves
5. Preparation of biopsy site with chlorhexidine swab
6. Application of sterile drapes
7. Local anaesthetic infiltration of biopsy site with 25G sterile needle (1ml of 2% lidocaine) – dispose of 25G needle in sharps bin
8. 3-mm disposable circular punch blade, scissors and forceps used to obtain skin biopsy to include dermis and epidermis
9. Immediately dispose of circular blade
10. Apply sterile gauze over biopsy site and apply pressure for 2 minutes
11. Steristrips used to close the incision. If haemostasis is difficult to achieve, use one suture (and give patient information about suture removal).
12. Apply waterproof dressing to biopsy site once haemostasis has been achieved.

Tissue Processing

1. Using sterile scissors or scalpel blade, and holding the biopsy with forceps, cut the biopsy approximately in half to produce sample A and sample B

Sample A:

- a. Place in cold Zamboni fixative (recipe below). This will also inactivate HIV, if biopsy is derived for the study of neuropathies associated with the virus or its treatment.
- b. Store in locked laboratory overnight. Specimen pot labelled as biohazard and with ID number. After this step, HIV is no longer active and does not require special measures.
- c. Rinse sample with 0.1M phosphate buffer
- d. Store at 4°C in cryoprecipitant (20% sucrose in 0.1M phosphate buffer) for 24 hours. Leak proof pot is double contained in a box labelled as a biohazard.
- e. Embed samples in cassettes or eppendorfs containing OCT
- f. Store in locked freezer at -80°C double contained and labelled as biohazard
- g. This sample will be used for immunohistochemistry

Sample B:

- a. Place in Eppendorf and snap freeze with either the liquid nitrogen method, or by immersing the Eppendorf in isopentane cooled with dry ice (once no longer effervescing isopentane at the correct temperature).
- b. Double contain and store in locked -80°C freezer and label as biohazard

Recipe For Making Zamboni (can also be bought ready-made commercially)

1. Make up aliquots of 16% paraformaldehyde
2. Make up 0.1M phosphate buffer solution
3. Combine 125mL paraformaldehyde with 150mL saturated aqueous picric acid
4. Make the combined paraformaldehyde-picric acid solution up to 1L with 0.1M phosphate buffer.
5. Adjust to pH 7.3 with sodium hydroxide
6. Once made, Zamboni solution is stable at 15-30 degrees C for up to 1 year.